209650: metagranodiorite vein, Success Well

(Cundimurra Monzogranite, Tuckanarra Suite, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Cue (SG 50-15), Austin (2442)
MGA Zone 50, 585333E 6911764N

Sampled on 10 October 2012

This sample was collected from a low outcrop on Wynyangoo Station, about 4.5 km west of Deep Well, 3.6 km northwest of Carey Well, and 3.1 km east-northeast of Success Well.

Geological context

The unit sampled is the Cundimurra Monzogranite of the Tuckanarra Suite (Van Kranendonk et al., 2013; Zibra et al., 2014; Zibra, 2020). The 2697–2657 Ma Cundimurra Monzogranite is a north-trending, elongate pluton, up to 200 km long and 25 km wide, which consists mainly of metamonzogranite and metasyenogranite, but also includes metatonalite, leucocratic muscovite metamonzogranite, and minor quartz metadiorite veins (Zibra, 2020). Plutons of the Tuckanarra Suite consist of strongly foliated and locally magmatically layered metagranodiorite to metamonzogranite. Tuckanarra Suite metagranitic rocks intrude the Murchison Supergroup and are crosscut by non-deformed granitic plutons of the Bald Rock Supersuite (Van Kranendonk et al. 2013; Zibra, 2020). The metagranodiorite vein at this locality is intruded into an older metamonzogranite component of the Cundimurra Monzogranite, a sample of which yielded an igneous crystallization age of 2698 ± 7 Ma (GSWA 209651, Wingate et al., 2021). A monzogranite dyke, intruded into the c. 2724 Ma Crossroads Granodiorite (Zibra, 2019) about 47 km to the northwest, yielded an igneous crystallization age of 2684 ± 3 Ma and is assigned to the Tuckanarra Suite (GSWA 155514, Lu et al., 2017). Samples of the Cundimurra Monzogranite, collected about 130 and 144 km to the south, yielded igneous crystallization ages of 2679 ± 8 Ma (GSWA 185997, Wingate et al., 2015a) and 2681 ± 5 Ma (GSWA 198212, Wingate et al., 2015b), respectively.

Petrographic description

The sample is an inequigranular, fine- to medium-grained metagranodiorite (Fig. 1), consisting of about 40% plagioclase, 20% quartz, 17% K-feldspar, 12% biotite, 5% recrystallized aggregates, 2–3% myrmekite, 2% magnetite, 1% titanite and accessory allanite, apatite, chlorite and zircon. Plagioclase is subhedral to euhedral, tabular, up to 3 mm long, and some crystals exhibit compositional zoning. Quartz forms anhedral, irregular, strained grains up to 1.5 mm in size, as well as irregular aggregates of less strained smaller grains. K-feldspar is anhedral to subhedral, mainly equant, and up to 2 mm in size. The crystals are strained and some display fine microcline twinning. Irregular and smoothly curved lobes of myrmekite are common at the margins of K-feldspar. Recrystallized quartz–feldspar±biotite aggregates are up to 3 mm in size. Biotite is anhedral to subhedral, up to 1.5 mm long, and mostly in aggregates of non-aligned laths. Disseminated subhedral grains of magnetite up to 0.4 mm in diameter exhibit ilmenite exsolution lamellae. Titanite is anhedral, irregular, up to 0.7 mm in size, and partly encloses quartz and feldspar. Subhedral, partly metamict allanite is up to 0.2 mm across, and euhedral laths of apatite are 0.1 mm long. Feldspars are slightly altered to clay minerals, and there is weak sericite alteration of some plagioclase, and minor replacement of biotite by chlorite. The granodiorite lacks a tectonic foliation, although strained grains, microcline twinning, myrmekite development, and recrystallization suggest deformation and recovery at moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are pale brown to dark brown and subhedral to euhedral. The crystals are up to 300 μm long, and mostly elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, many crystals contain high-U, metamict zones, and some appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 30–31 July 2020, using SHRIMP-B. Twelve analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.22% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-three analyses were obtained from 23 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

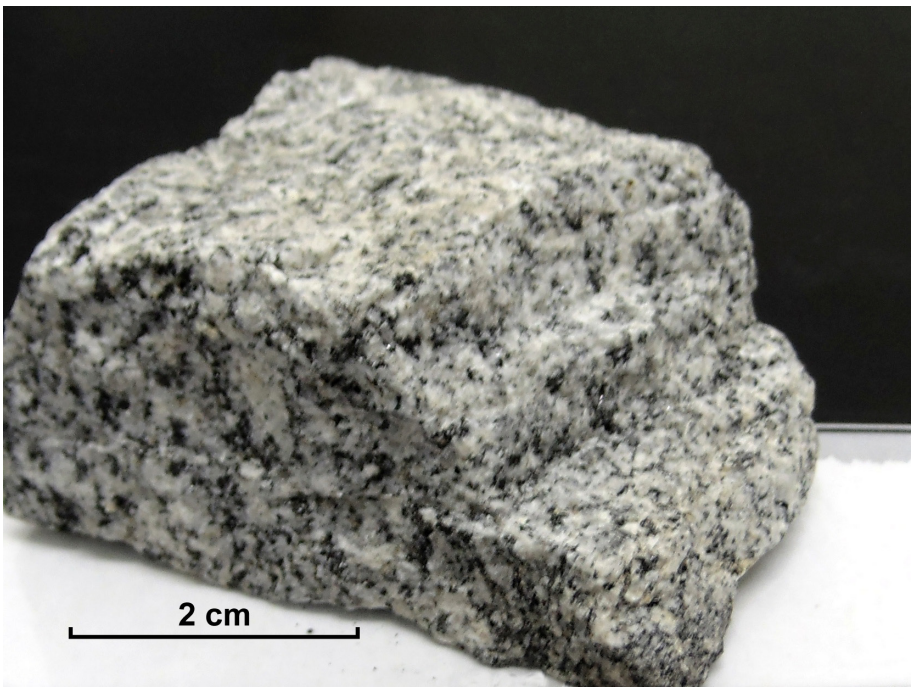

Figure 1. Hand specimen image of sample 209650: metagranodiorite vein, Success Well

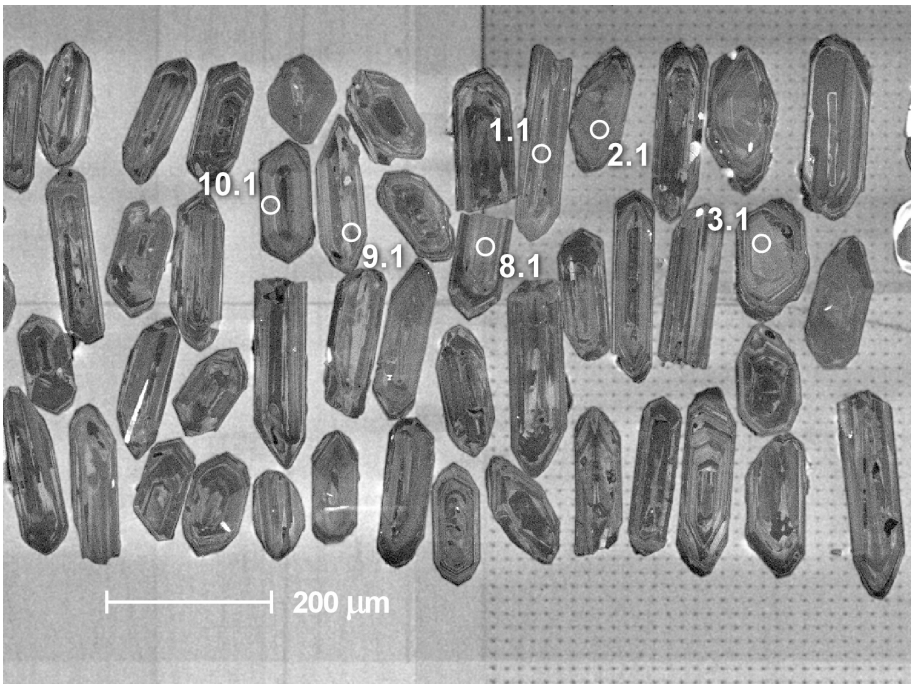

Figure 2. Cathodoluminescence image of representative zircons from sample 209650: metagranodiorite vein, Success Well. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 209650: metagranodiorite vein, Success Well

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	20	20.1	310	175	0.56	0.047	1.996	0.019	0.18250	0.00120	1.997	0.019	0.18209	0.00121	2618	21	2672	11	2.0
I	21	21.1	288	153	0.53	0.063	2.002	0.020	0.18274	0.00079	2.003	0.020	0.18217	0.00083	2610	22	2673	8	2.3
I	22	22.1	260	168	0.64	0.000	1.975	0.020	0.18247	0.00080	1.975	0.020	0.18247	0.00080	2641	22	2676	7	1.3
I	1	1.1	499	593	1.19	0.157	1.957	0.017	0.18405	0.00055	1.960	0.017	0.18265	0.00064	2658	19	2677	6	0.7
I	6	6.1	227	169	0.74	0.109	2.015	0.020	0.18388	0.00175	2.017	0.020	0.18291	0.00178	2595	22	2679	16	3.1
I	2	2.1	370	256	0.69	0.012	1.946	0.017	0.18311	0.00062	1.946	0.017	0.18300	0.00062	2673	20	2680	6	0.3
I	19	19.1	400	365	0.91	0.021	1.977	0.018	0.18367	0.00066	1.978	0.018	0.18348	0.00067	2638	20	2685	6	1.7
I	23	23.1	324	150	0.46	-0.018	1.959	0.019	0.18371	0.00073	1.959	0.019	0.18387	0.00074	2659	21	2688	7	1.1
I	9	9.1	389	356	0.92	0.026	1.998	0.018	0.18460	0.00150	1.998	0.018	0.18436	0.00151	2616	19	2693	13	2.9
I	12	12.1	593	651	1.10	0.019	1.969	0.016	0.18480	0.00055	1.969	0.016	0.18463	0.00056	2648	18	2695	5	1.8
I	10	10.1	128	74	0.58	0.151	2.032	0.027	0.18624	0.00236	2.035	0.027	0.18490	0.00243	2577	28	2697	22	4.5
I	15	15.1	299	145	0.48	-0.038	1.974	0.019	0.18504	0.00076	1.974	0.019	0.18537	0.00078	2643	21	2702	7	2.2
I	7	7.1	189	139	0.74	0.045	2.021	0.023	0.18687	0.00162	2.022	0.023	0.18647	0.00164	2590	24	2711	14	4.5
X	3	3.1	133	60	0.45	-0.022	1.975	0.025	0.18836	0.00122	1.975	0.025	0.18856	0.00123	2641	28	2730	11	3.2
X	14	14.1	204	145	0.71	0.366	1.945	0.021	0.19358	0.00094	1.952	0.021	0.19031	0.00114	2667	24	2745	10	2.9
X	11	11.1	142	94	0.66	0.447	1.930	0.024	0.20003	0.00112	1.938	0.024	0.19605	0.00139	2682	27	2794	12	4.0
X	17	17.1	304	284	0.94	0.008	1.935	0.019	0.19758	0.00076	1.935	0.019	0.19750	0.00076	2685	22	2806	6	4.3
D	8	8.1	1197	1092	0.91	0.031	2.406	0.044	0.17515	0.00040	2.407	0.044	0.17487	0.00041	2240	35	2605	4	14.0
D	13	13.1	393	360	0.92	0.082	2.128	0.019	0.17945	0.00066	2.129	0.019	0.17872	0.00070	2482	19	2641	6	6.0
D	18	18.1	304	231	0.76	0.696	3.769	0.156	0.18970	0.00093	3.795	0.157	0.18349	0.00140	1508	56	2685	13	43.8
D	5	5.1	120	62	0.52	0.383	2.051	0.026	0.19311	0.00117	2.059	0.027	0.18970	0.00143	2552	27	2740	12	6.8
D	16	16.1	334	123	0.37	0.228	2.060	0.021	0.19198	0.00084	2.065	0.021	0.18995	0.00095	2546	22	2742	8	7.1
D	4	4.1	162	51	0.31	-0.066	1.936	0.022	0.20881	0.00099	1.935	0.022	0.20939	0.00102	2685	25	2901	8	7.4

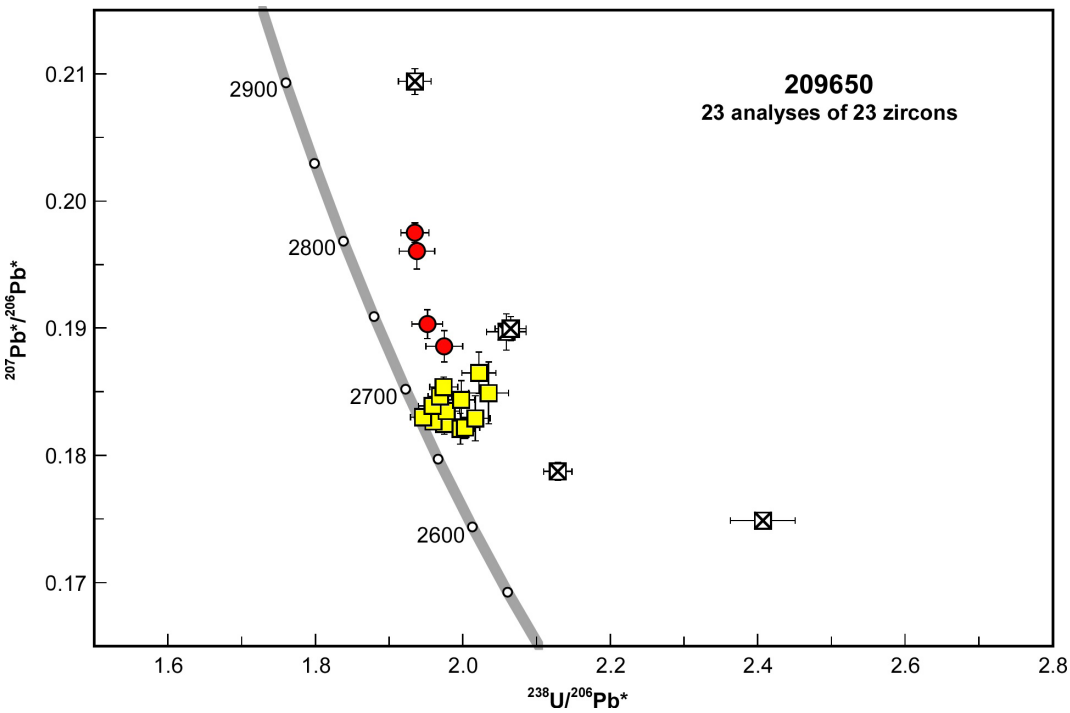

Figure 3. U–Pb analytical data for sample 209650: metagranodiorite vein, Success Well. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircon cores); crossed squares indicate Group D (discordance >5%). One highly discordant analysis in Group D is not shown

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Six analyses are >5% discordant. The dates obtained from these six analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 17 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios and their positions within the crystals.

Group I comprises 13 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2685 ± 6 Ma (MSWD = 1.8).

Group X comprises four analyses of zircon cores (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2806–2730 Ma.

The date of 2685 ± 6 Ma for the 13 analyses in Group I is interpreted as the igneous crystallization age of the granodiorite. The dates of 2806–2730 Ma for the four slightly discordant analyses in Group X are interpreted as the minimum ages of inherited zircons.

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Zibra, I 2021, 209650: metagranodiorite vein, Success Well; Geochronology Record 1830: Geological Survey of Western Australia, 4p.

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